

## Q1. EACH STUDENT WITH THEIR MENTOR NAME

### Problem Statement

CodeSphere Academy runs a student mentorship program where experienced students guide junior learners.

Each student record stores the ID of the mentor assigned to them. The academic team wants a report showing every student alongside their mentor's name to better visualize the mentorship structure.

Students who do not have a mentor (mentor\_id is NULL) should not appear in the report.

### Task

Write an SQL query that joins the students table to itself and returns every student along with the name of their mentor.

The output must contain the following columns:

- student\_name — Name of the student
- mentor\_name — Name of the student's mentor

The results should be sorted alphabetically by student\_name.

### Input Format

#### TABLE SCHEMA

Table: students

| Column Name  | Data Type    | Description               |
|--------------|--------------|---------------------------|
| student_id   | INT          | Unique student identifier |
| student_name | VARCHAR(100) | Student name              |
| mentor_id    | INT          | Mentor student ID         |
| course       | VARCHAR(50)  | Course enrolled           |

### Database Creation Script

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(100),  
    mentor_id INT,  
    course VARCHAR(50)  
);
```

### Sample Data

```
INSERT INTO students VALUES  
(1,'Arun',NULL,'Python'),  
(2,'Priya',1,'Java'),  
(3,'Rahul',1,'SQL'),  
(4,'Sneha',2,'Python'),  
(5,'Vikram',2,'Java'),  
(6,'Neeraj',3,'SQL');
```

### Sample Input

The students table contains:

| student_id | student_name | mentor_id | course |
|------------|--------------|-----------|--------|
| 1          | Arun         | NULL      | Python |
| 2          | Priya        | 1         | Java   |
| 3          | Rahul        | 1         | SQL    |
| 4          | Sneha        | 2         | Python |
| 5          | Vikram       | 2         | Java   |

### Output Format

Return the following columns in exact order:

student\_name mentor\_name

### Sample Output

| student_name | mentor_name |
|--------------|-------------|
| Priya        | Arun        |
| Rahul        | Arun        |
| Sneha        | Priya       |
| Vikram       | Priya       |

### Explanation

Among the sample rows:

- Arun has mentor\_id = NULL and therefore has no mentor.
- Arun is excluded from the output.
- Priya and Rahul are mentored by Arun.
- Sneha and Vikram are mentored by Priya.

The report displays each student alongside the name of their mentor.

### Constraints

- Keywords, table names, column names, and record values are case-sensitive.
- All records are pre-populated.
- No console input is required.
- Use exact output column names:
  - student\_name
  - mentor\_name
- Students with mentor\_id = NULL must not appear.
- Results must be ordered by student\_name ascending.

### Final Answer

```
SELECT
  s.student_name,
  m.student_name AS mentor_name
FROM students s
INNER JOIN students m
  ON s.mentor_id = m.student_id
ORDER BY s.student_name;
```

## EACH EMPLOYEE WITH THEIR TEAM LEAD

### Problem Statement

TechNova Solutions organizes employees into project teams. Every employee reports to a team lead who is also an employee in the organization.

The HR department wants to generate a report that shows each employee together with the name of their team lead.

Employees who are themselves team leads and do not report to anyone should not appear in the report.

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### Task

Write an SQL query that joins the employees table to itself and returns every employee along with the name of their team lead.

The output must contain the following columns:

- employee\_name — Name of the employee
- lead\_name — Name of the team lead

Sort the result alphabetically by employee\_name.

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### Input Format

#### TABLE SCHEMA

Table: employees

| Column Name   | Data Type    | Description           |
|---------------|--------------|-----------------------|
| employee_id   | INT          | Employee ID           |
| employee_name | VARCHAR(100) | Employee Name         |
| lead_id       | INT          | Team Lead Employee ID |
| project_name  | VARCHAR(100) | Assigned Project      |

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### Database Creation Script

```
CREATE TABLE employees (  
    employee_id INT PRIMARY KEY,  
    employee_name VARCHAR(100),  
    lead_id INT,  
    project_name VARCHAR(100)  
);
```

---

### Sample Data

```
INSERT INTO employees VALUES  
(1,'Raj',NULL,'ERP System'),  
(2,'Amit',1,'ERP System'),  
(3,'Neha',1,'ERP System'),  
(4,'Karan',2,'CRM Portal'),  
(5,'Divya',2,'CRM Portal'),  
(6,'Rohit',3,'Mobile App');
```

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### Sample Input

| employee_id | employee_name | lead_id | project_name |
|-------------|---------------|---------|--------------|
| 1           | Raj           | NULL    | ERP System   |
| 2           | Amit          | 1       | ERP System   |
| 3           | Neha          | 1       | ERP System   |
| 4           | Karan         | 2       | CRM Portal   |
| 5           | Divya         | 2       | CRM Portal   |

### Output Format

Return the following columns in exact order:

employee\_name

lead\_name

### Sample Output

| employee_name | lead_name |
|---------------|-----------|
| Amit          | Raj       |
| Divya         | Amit      |
| Karan         | Amit      |
| Neha          | Raj       |

### Explanation

- Raj has no team lead (lead\_id = NULL).
- Raj is excluded from the report.
- Amit and Neha report to Raj.
- Karan and Divya report to Amit.

### Constraints

- Keywords, table names, column names, and record values are case-sensitive.
- All records are pre-populated.
- No console input is required.
- Use exact output column names:
  - employee\_name
  - lead\_name
- Employees with lead\_id IS NULL must not appear.
- Results must be ordered by employee\_name ascending.

### Final Answer

```
SELECT
  e.employee_name,
  l.employee_name AS lead_name
FROM employees e
INNER JOIN employees l
  ON e.lead_id = l.employee_id
ORDER BY e.employee_name;
```

## EACH DOCTOR WITH THEIR SENIOR DOCTOR

### problem Statement

CityCare Hospital maintains a hierarchical medical structure where junior doctors work under senior doctors.

Each doctor record stores the ID of the senior doctor supervising them.

The administration wants a report showing every doctor together with the name of their senior doctor.

### Task

Write an SQL query that returns every doctor and their supervising senior doctor.

The output must contain:

- doctor\_name
- senior\_doctor

### Input Format

#### TABLE SCHEMA

Table: doctors

| Column Name    | Data Type    |
|----------------|--------------|
| doctor_id      | INT          |
| doctor_name    | VARCHAR(100) |
| senior_id      | INT          |
| specialization | VARCHAR(50)  |

### Database Creation Script

```
CREATE TABLE doctors(  
    doctor_id INT PRIMARY KEY,  
    doctor_name VARCHAR(100),  
    senior_id INT,  
    specialization VARCHAR(50)  
);
```

### Sample Data

```
INSERT INTO doctors VALUES  
(1,'Dr. Rao',NULL,'Cardiology'),  
(2,'Dr. Khan',1,'Cardiology'),  
(3,'Dr. Shah',1,'Neurology'),  
(4,'Dr. Mehta',2,'Cardiology'),  
(5,'Dr. Patel',3,'Neurology');
```

### Sample Output

| doctor_name | senior_doctor |
|-------------|---------------|
| Dr. Khan    | Dr. Rao       |
| Dr. Mehta   | Dr. Khan      |
| Dr. Patel   | Dr. Shah      |
| Dr. Shah    | Dr. Rao       |

### Explanation

- Dr. Rao is the head doctor and has no senior.
- Dr. Khan and Dr. Shah report to Dr. Rao.
- Dr. Mehta reports to Dr. Khan.
- Dr. Patel reports to Dr. Shah.

### Constraints

- Use exact column names:
  - doctor\_name
  - senior\_doctor
- Doctors with senior\_id IS NULL must not appear.
- Order by doctor\_name.

### Final Answer

```
SELECT
    d.doctor_name,
    s.doctor_name AS senior_doctor
FROM doctors d
INNER JOIN doctors s
    ON d.senior_id = s.doctor_id
ORDER BY d.doctor_name;
```

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## EACH PLAYER WITH THEIR COACH

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### Problem Statement

A national cricket academy maintains player records and coaching assignments. Every player is assigned a coach who is also listed in the players table.

The management wants a report displaying each player along with the name of their coach. Players who do not have a coach assigned should not appear in the report.

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### Task

Write an SQL query that joins the players table to itself and returns every player along with the name of their coach.

The output must contain:

- player\_name
  - coach\_name
- 

### TABLE SCHEMA

#### Table: players

| Column Name | Data Type    |
|-------------|--------------|
| player_id   | INT          |
| player_name | VARCHAR(100) |
| coach_id    | INT          |
| role        | VARCHAR(50)  |

---

### Database Creation Script

```
CREATE TABLE players(  
  player_id INT PRIMARY KEY,  
  player_name VARCHAR(100),  
  coach_id INT,  
  role VARCHAR(50)  
);
```

---

### Sample Data

```
INSERT INTO players VALUES  
(1,'Virat',NULL,'Coach'),  
(2,'Rohit',1,'Batsman'),  
(3,'Gill',1,'Batsman'),  
(4,'Ishan',2,'Wicket Keeper'),  
(5,'Hardik',2,'All Rounder');
```

---

### Sample Output

| player_name | coach_name |
|-------------|------------|
| Gill        | Virat      |
| Hardik      | Rohit      |
| Ishan       | Rohit      |
| Rohit       | Virat      |

---

### Final Answer

```
SELECT  
  p.player_name,  
  c.player_name AS coach_name  
FROM players p  
JOIN players c  
ON p.coach_id = c.player_id  
ORDER BY p.player_name;
```

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## EACH SALES REPRESENTATIVE WITH THEIR MANAGER

### Problem Statement

ABC Sales Corporation maintains a hierarchy where sales representatives report to managers. Management wants a report listing every sales representative along with their manager's name.

### TABLE SCHEMA

Table: sales\_reps

| Column Name | Data Type    |
|-------------|--------------|
| rep_id      | INT          |
| rep_name    | VARCHAR(100) |
| manager_id  | INT          |
| region      | VARCHAR(50)  |

### Database Creation Script

```
CREATE TABLE sales_reps(  
  rep_id INT PRIMARY KEY,  
  rep_name VARCHAR(100),  
  manager_id INT,  
  region VARCHAR(50)  
);
```

### Sample Data

```
INSERT INTO sales_reps VALUES  
(1,'Ravi',NULL,'North'),  
(2,'Mohan',1,'North'),  
(3,'Geeta',1,'South'),  
(4,'Anita',2,'North');
```

### Sample Output

| rep_name | manager_name |
|----------|--------------|
| Anita    | Mohan        |
| Geeta    | Ravi         |
| Mohan    | Ravi         |

### Final Answer

```
SELECT  
  r.rep_name,  
  m.rep_name AS manager_name  
FROM sales_reps r  
JOIN sales_reps m  
ON r.manager_id = m.rep_id  
ORDER BY r.rep_name;
```

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## EACH TEACHER WITH THEIR HOD

---

### Problem Statement

A university stores teacher records and department hierarchy information.

Each teacher reports to a Head of Department (HOD).

Generate a report displaying every teacher along with the name of their HOD.

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### TABLE SCHEMA

Table: teachers

| Column Name  | Data Type    |
|--------------|--------------|
| teacher_id   | INT          |
| teacher_name | VARCHAR(100) |
| hod_id       | INT          |
| department   | VARCHAR(50)  |

---

### Database Creation Script

```
CREATE TABLE teachers(  
teacher_id INT PRIMARY KEY,  
teacher_name VARCHAR(100),  
hod_id INT,  
department VARCHAR(50)  
);
```

---

### Sample Data

```
INSERT INTO teachers VALUES  
(1,'Prof. Sharma',NULL,'Computer Science'),  
(2,'Prof. Gupta',1,'Computer Science'),  
(3,'Prof. Jain',1,'Computer Science'),  
(4,'Prof. Verma',2,'Computer Science');
```

---

### Sample Output

| teacher_name | hod_name     |
|--------------|--------------|
| Prof. Gupta  | Prof. Sharma |
| Prof. Jain   | Prof. Sharma |
| Prof. Verma  | Prof. Gupta  |

---

### Final Answer

```
SELECT  
    t.teacher_name,  
    h.teacher_name AS hod_name  
FROM teachers t  
JOIN teachers h  
ON t.hod_id = h.teacher_id  
ORDER BY t.teacher_name;
```

## EACH AGENT WITH THEIR SUPERVISOR

### Problem Statement

A customer service company tracks agents and supervisors in the same table.

Each agent reports to a supervisor.

Generate a report showing every agent and their supervisor.

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### TABLE SCHEMA

Table: agents

| Column Name   | Data Type    |
|---------------|--------------|
| agent_id      | INT          |
| agent_name    | VARCHAR(100) |
| supervisor_id | INT          |
| city          | VARCHAR(50)  |

---

### Database Creation Script

```
CREATE TABLE agents(  
  agent_id INT PRIMARY KEY,  
  agent_name VARCHAR(100),  
  supervisor_id INT,  
  city VARCHAR(50)  
);
```

---

### Sample Data

```
INSERT INTO agents VALUES  
(1,'Rajesh',NULL,'Mumbai'),  
(2,'Kunal',1,'Mumbai'),  
(3,'Ritika',1,'Delhi'),  
(4,'Aman',2,'Pune');
```

---

### Final Answer

```
SELECT  
  a.agent_name,  
  s.agent_name AS supervisor_name  
FROM agents a  
JOIN agents s  
ON a.supervisor_id = s.agent_id  
ORDER BY a.agent_name;
```

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## EACH FRANCHISE WITH ITS PARENT FRANCHISE

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### Problem Statement

A food chain stores franchise hierarchy information.  
Every branch franchise points to its parent franchise.  
Display each franchise with its parent franchise.

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### TABLE SCHEMA

Table: franchises

| Column Name    | Data Type    |
|----------------|--------------|
| franchise_id   | INT          |
| franchise_name | VARCHAR(100) |
| parent_id      | INT          |

---

### Database Creation Script

```
CREATE TABLE franchises(  
franchise_id INT PRIMARY KEY,  
franchise_name VARCHAR(100),  
parent_id INT  
);
```

---

### Sample Data

```
INSERT INTO franchises VALUES  
(1,'Global Foods',NULL),  
(2,'Global Foods Mumbai',1),  
(3,'Global Foods Pune',1),  
(4,'Global Foods Nashik',2);
```

---

### Final Answer

```
SELECT  
    f.franchise_name,  
    p.franchise_name AS parent_franchise  
FROM franchises f  
JOIN franchises p  
ON f.parent_id = p.franchise_id  
ORDER BY f.franchise_name;
```

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## EACH DEPARTMENT WITH ITS PARENT DEPARTMENT

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### Problem Statement

A multinational company stores department hierarchy information.

Each department may belong to a parent department.

Generate a report showing every department along with its parent department.

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### TABLE SCHEMA

Table: departments

| Column Name          | Data Type    |
|----------------------|--------------|
| department_id        | INT          |
| department_name      | VARCHAR(100) |
| parent_department_id | INT          |

---

### Database Creation Script

```
CREATE TABLE departments(  
  department_id INT PRIMARY KEY,  
  department_name VARCHAR(100),  
  parent_department_id INT  
);
```

---

### Sample Data

```
INSERT INTO departments VALUES  
(1,'Head Office',NULL),  
(2,'Sales',1),  
(3,'HR',1),  
(4,'Recruitment',3),  
(5,'Training',3);
```

---

### Sample Output

| department_name | parent_department |
|-----------------|-------------------|
| HR              | Head Office       |
| Recruitment     | HR                |
| Sales           | Head Office       |
| Training        | HR                |

---

### Final Answer

```
SELECT  
  d.department_name,  
  p.department_name AS parent_department  
FROM departments d  
JOIN departments p  
ON d.parent_department_id = p.department_id  
ORDER BY d.department_name;
```

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## EACH ENGINEER WITH THEIR REPORTING MANAGER

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### Problem Statement

A software company maintains engineer reporting structures in a single table.

Each engineer reports to a manager who is also an engineer.

Generate a report showing every engineer and their reporting manager.

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### TABLE SCHEMA

Table: engineers

| Column Name   | Data Type    |
|---------------|--------------|
| engineer_id   | INT          |
| engineer_name | VARCHAR(100) |
| manager_id    | INT          |
| technology    | VARCHAR(50)  |

---

### Database Creation Script

```
CREATE TABLE engineers(  
  engineer_id INT PRIMARY KEY,  
  engineer_name VARCHAR(100),  
  manager_id INT,  
  technology VARCHAR(50)  
);
```

---

### Sample Data

```
INSERT INTO engineers VALUES  
(1,'Akash',NULL,'Java'),  
(2,'Nitin',1,'Java'),  
(3,'Pooja',1,'Python'),  
(4,'Sonal',2,'Java');
```

---

### Final Answer

```
SELECT  
  e.engineer_name,  
  m.engineer_name AS manager_name  
FROM engineers e  
JOIN engineers m  
ON e.manager_id = m.engineer_id  
ORDER BY e.engineer_name;
```